Aiden Emanuel Fairman

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EDUCATION

University of Bern **09/2024 - 08/2026**
M.S. Bioinformatics and Computational Biology
Boston University **09/2019 - 08/2023**
B.A. Computer Science

WORK EXPERIENCE

Novo Nordisk
Computational Biology Consultant - via Kelly Services | Remote **08/2025 - Present**

- Designed and maintained modular, process-oriented analytical workflows adaptable to changing project requirements.
- Coordinated with biologists, data scientists, and project stakeholders to align objectives, timelines, and deliverables.
- Built data processing and modeling pipelines in Python to transform large, noisy datasets into actionable insights.
- Translated complex technical analyses into clear, decision-ready summaries, enabling informed prioritization and next steps.
- Managed multiple parallel projects under tight deadlines in a highly regulated pharma R&D environment.

Novo Nordisk
Computational Biology Intern | Boston, MA **06/2025 - 08/2025**

- Evaluated and compared external technology platforms, analyzing performance trade-offs to support informed selection decisions.
- Processed and standardized high-dimensional datasets into analysis-ready, reusable formats, improving accessibility for downstream users.
- Communicated progress, risks, and recommendations to stakeholders while managing multiple workstreams in parallel.

Center for Neurological Imaging at Brigham and Women's Hospital - Harvard University Affiliated Laboratory
Research Trainee | Boston, MA **03/2024 - 10/2024**

- Developed a user-facing data exploration tool to enable non-technical stakeholders to interact with complex datasets.
- Led evaluation and selection of technologies and system designs, balancing usability, scalability, and integration constraints.
- Coordinated inputs from researchers, engineers, and end users to align system functionality with project needs.
- Designed structured workflows to integrate data from multiple sources into a unified platform.

MITRE
Software Engineer Intern | Bedford, MA **05/2022 - 08/2022**

- Contributed to a cross-functional Agile team delivering a minimum viable product under tight timelines and evolving requirements.
- Collaborated with engineers, project leads, and stakeholders to translate high-level needs into concrete deliverables.
- Supported the transition of early prototypes into stable, maintainable systems through structured development practices.
- Gained experience working in a high-accountability environment with strong emphasis on documentation and delivery.

PROJECT & RESEARCH

M.S. Thesis: Morphology-to-Transcriptome Prediction in CRC via Spatial Transcriptomics | University of Bern

- Investigating links between cellular morphology and gene expression state in colorectal cancer (CRC) tissue
- Analyzing single-cell spatial transcriptomics data with Scanpy and Squidpy to characterize morphology–expression co-variation
- Developing a supervised ML classifier (PyTorch) to predict transcriptomic profiles from histological cell morphology features

ClimateQA: Multi-Model LLM Benchmark for Climate Science | github.com/aidenfairman12 | aidenfairman.vercel.app

- Constructed a 172-question multiple-choice benchmark grounded in IPCC AR6 source passages, using four frontier LLMs (Claude, GPT-4o, Gemini, Llama) to generate and evaluate 580 candidate questions
- Designed evaluation methodology to analyse self vs. cross-model accuracy, surfacing systematic biases in how models perform on questions they generated versus questions from other models

FreightSignal: RAG-Powered Supply Chain Intelligence | github.com/aidenfairman12 | aidenfairman.vercel.app

- Built a supply chain disruption intelligence system using retrieval-augmented generation (RAG) on a live corpus of logistics news
- Implemented a retrieve-then-rerank pipeline (bi-encoder first pass → cross-encoder reranker) and evaluated end-to-end quality using RAGAS across 50 synthetic QA pairs

ACTIVITIES & AWARDS

NCAA Division I Athlete | Boston University Swim & Dive Team

- Demonstrated discipline, teamwork, and performance under pressure in a highly competitive environment.

Food Pantry Volunteer | Boston Cares

- Volunteer at various food pantries in the Greater Boston Area through Boston Cares.

Global Engineering Brigades Member | Boston University

- Local school chapter of international nonprofit that partners with under resourced communities to help establish clean water pipe systems.

LANGUAGE SKILLS

English - Native
French - B1

SKILLS

- **Machine Learning:** Supervised and deep learning (classification, regression), model training and evaluation, feature engineering, PyTorch; statistical learning fundamentals
- **Bioinformatics & Computational Biology:** Single-cell and spatial transcriptomics (Scanpy, Squidpy), multi-modal biological data processing, bioinformatics pipeline design
- **Workflow & Collaboration:** Agile/Scrum, stakeholder communication, cross-functional teams, decision-support presentations, parallel workload management
- **Analytical & Technical Skills:** Data analysis and visualization, modular systems and pipeline-based architectures, workflow design andstandardization
- **Programming:** Python, R, Bash, SQL, JavaScript
- **Other Technical:** Git, Full-stack Web Dev (JS, HTML/CSS, RESTful APIs), Bioinformatics tools

References

Available upon request.